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The State of Open-Source Large Language Models:

14 min read · Mar 1, 2026

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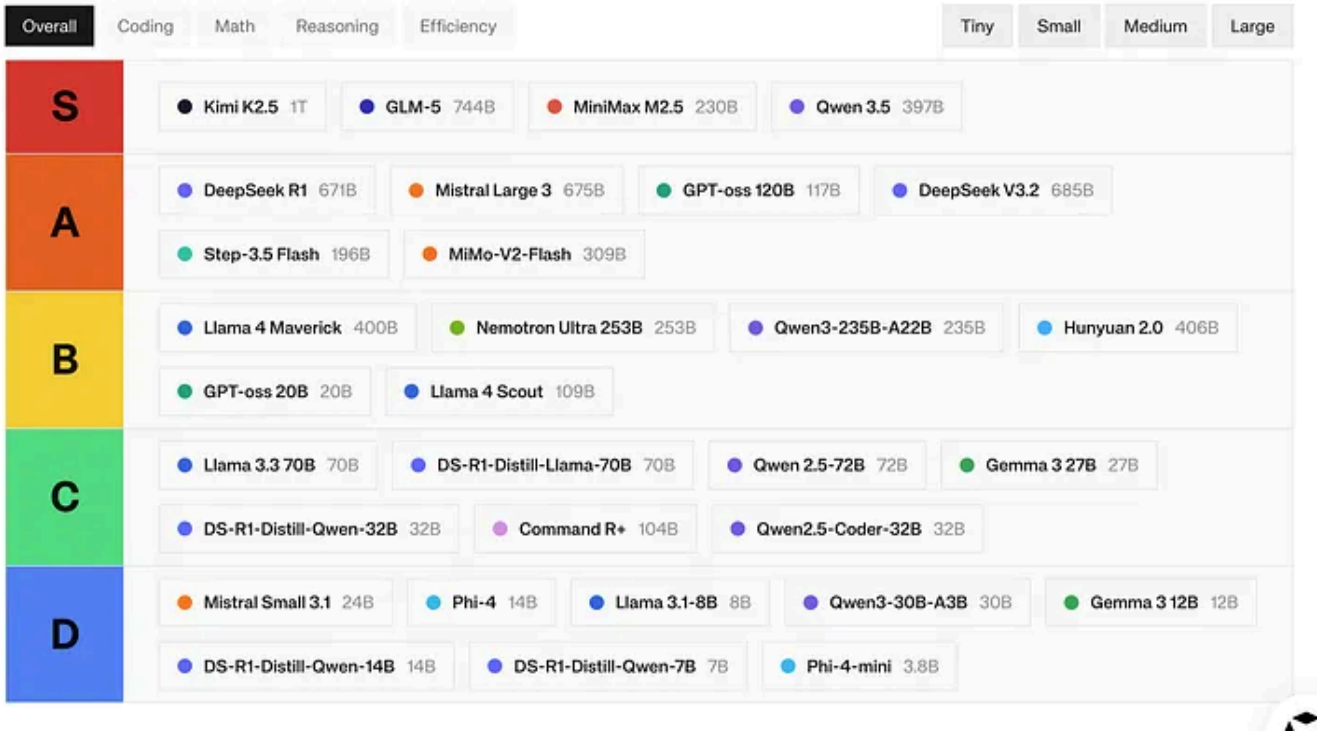
A Comprehensive Tier List Review for 2026,

An in-depth analysis of how open-weight models are reshaping the AI landscape



Introduction: The Open-Source AI Revolution

The artificial intelligence landscape has undergone a seismic transformation over the past two years. What was once the exclusive domain of a handful of well-funded laboratories — OpenAI, Google DeepMind, and Anthropic — has become a vibrant, fiercely competitive ecosystem driven by open-source and open-weight models. The

latest overall tier list ranking of open-source large language models, presented in a now-viral infographic that categorizes models from S-tier down to D-tier, reveals an industry that has matured at a breathtaking pace. Models from Chinese AI labs, European startups, American tech giants, and even hardware manufacturers now compete for supremacy across multiple performance categories including overall quality, coding, math, reasoning, and efficiency.

Tier	Model	Developer	Parameters	Launch Date
S	Kimi K2.5	Moonshot AI	1T (32B active)	January 27, 2026
S	GLM-5	Zhipu AI (Z.ai)	744B (40B active)	February 11, 2026
S	MiniMax M2.5	MiniMax	230B (10B active)	February 12, 2026
S	Qwen 3.5	Alibaba Cloud	397B (17B active)	February 16, 2026
A	DeepSeek R1	DeepSeek	671B (37B active)	January 20, 2025
A	Mistral Large 3	Mistral AI	675B (41B active)	December 2, 2025
A	GPT-oss 120B	OpenAI	117B (5.1B active)	August 5, 2025
A	DeepSeek V3.2	DeepSeek	685B (37B active)	December 1, 2025
A	Step-3.5 Flash	StepFun	196B	Q1 2026
A	MiMo-V2-Flash	Xiaomi	309B	Q1 2026
B	Llama 4 Maverick	Meta	400B	April 5, 2025
B	Nemotron Ultra 253B	NVIDIA	253B	March 18, 2025
B	Qwen3-235B-A22B	Alibaba Cloud	235B (22B active)	April 29, 2025
B	Hunyuan 2.0	Tencent	406B	Q4 2025
B	GPT-oss 20B	OpenAI	21B (3.6B active)	August 5, 2025
B	Llama 4 Scout	Meta	109B	April 5, 2025
C	Llama 3.3 70B	Meta	70B	December 6, 2024
C	DS-R1-Distill-Llama-70B	DeepSeek	70B	January 20, 2025
C	Qwen 2.5-72B	Alibaba Cloud	72B	September 19, 2024
C	Gemma 3 27B	Google	27B	March 12, 2025
C	DS-R1-Distill-Qwen-32B	DeepSeek	32B	January 20, 2025
C	Command R+	Cohere	104B	April 4, 2024
C	Qwen2.5-Coder-32B	Alibaba Cloud	32B	November 12, 2024
D	Mistral Small 3.1	Mistral AI	24B	March 18, 2025
D	Phi-4	Microsoft	14B	February 26, 2025
D	Llama 3.1-8B	Meta	8B	July 23, 2024
D	Qwen3-30B-A3B	Alibaba Cloud	30B (3B active)	April 29, 2025
D	Gemma 3 12B	Google	12B	March 12, 2025
D	DS-R1-Distill-Qwen-14B	DeepSeek	14B	January 20, 2025
D	DS-R1-Distill-Qwen-7B	DeepSeek	7B	January 20, 2025
D	Phi-4-mini	Microsoft	3.8B	February 26, 2025

Note: "Active" parameters refer to MoE models where only a fraction of total parameters are used per forward pass. Q1 2026 indicates approximate release windows where exact dates are unconfirmed.

This review provides an exhaustive analysis of the current open-source LLM landscape as captured by this tier list. We examine not only where each model falls within the ranking system, but what these placements mean for developers, researchers, enterprises, and the broader trajectory of artificial intelligence. We explore the technical innovations driving each tier, the geopolitical implications of model development patterns, the economic considerations for deployment, and the emerging trends that will shape the next generation of open-source AI.

Before diving into the tier-by-tier analysis, it is worth noting what makes this particular moment so significant. For the first time in the history of AI, the best open-source models are genuinely competitive with proprietary frontier systems across most benchmarks. The gap that once seemed insurmountable — between what you could access freely and what required expensive API subscriptions — has narrowed to the point where many organizations can now build world-class AI products entirely on open foundations. This is not merely a technical achievement; it is a fundamental shift in the economics and accessibility of artificial intelligence.

Complete Model Reference: Tier, Parameters & Launch Dates

The following table provides a comprehensive reference of all models in the tier list, including their developers, parameter counts, and official launch dates. Notably, the entire S-tier was released within a three-week window in late January to mid-February 2026, illustrating the extraordinary pace of competition among leading AI labs.

efer to MoE models where only a fraction of total parameters are used per forward pass. Q1 2026 indicates approximate release windows where exact dates are unconfirmed.

S-Tier: The Open-Source Elite

The S-tier represents the absolute pinnacle of open-source model performance — the rare models that can genuinely stand alongside or even surpass some proprietary offerings in head-to-head evaluations. Only four models have earned this distinction: **Kimi K2.5 (1T)**, **GLM-5 (744B)**, **MiniMax M2.5 (230B)**, and **Qwen 3.5 (397B)**. Remarkably, all four were released within a three-week span between January 27 and February 16, 2026 — a period that will likely be remembered as the watershed moment for open-source AI.

Kimi K2.5 — The Trillion-Parameter Titan (January 27, 2026)

Kimi K2.5 from Moonshot AI stands as the largest model on the entire list at a staggering 1 trillion parameters, with 32 billion active per forward pass via its MoE

architecture. Launched on January 27, 2026, it was the opening salvo of what became an extraordinary week of Chinese model releases. Moonshot AI has combined massive pretraining on approximately 15 trillion mixed visual and text tokens with sophisticated post-training alignment, including extensive reinforcement learning. The result is a model that does not just know more but reasons more coherently, follows instructions more precisely, and generates more nuanced responses than any other open-weight model available.

Its placement at the top of the S-tier suggests that scale, when combined with high-quality data and thoughtful training methodology, still matters enormously. There has been much debate in the AI community about whether we have hit the scaling wall. Kimi K2.5's performance provides a definitive counterargument: scale continues to unlock emergent capabilities, particularly in complex multi-step reasoning, nuanced language understanding, and cross-domain knowledge integration. The model also introduces an innovative Agent Swarm mechanism, enabling it to self-direct up to 100 sub-agents executing parallel workflows across up to 1,500 tool calls — cutting execution time by up to 4.5x compared to single-agent setups.

GLM-5 — Zhipu's Crown Jewel (February 11, 2026)

GLM-5 from Zhipu AI (now branded as Z.ai), at 744 billion parameters with 40 billion active, arrived on February 11, 2026 — just days before the Lunar New Year. What makes GLM-5's story especially compelling is that it was trained entirely on 100,000 Huawei Ascend 910B chips without a single NVIDIA GPU, making it a geopolitical milestone as much as a technical one. Zhipu had gone public on the Hong Kong Stock Exchange just five weeks earlier on January 8, raising $558 million — becoming the world's first publicly traded foundation model company.

The model's balance is what sets it apart: it delivers frontier-level results across coding, mathematics, reasoning, and general language tasks simultaneously. It scored 50.4% on Humanity's Last Exam with tools — outperforming both Claude Opus 4.5 and GPT-5.2. Its novel "Slime" asynchronous reinforcement learning framework compressed its hallucination rate dramatically, achieving the industry's best score on the Artificial Analysis Omniscience Index. For enterprise applications requiring long-document analysis and multi-turn conversational contexts, GLM-5's 200K-token context window and DeepSeek Sparse Attention mechanism offer capabilities that many proprietary alternatives struggle to match.

MiniMax M2.5 — The Efficiency Champion (February 12, 2026)

Perhaps the most impressive story in S-tier belongs to MiniMax M2.5, released on February 12, 2026 — just one day after GLM-5. At 230 billion total parameters with only 10 billion active per forward pass, it is the smallest model in the S-tier by a significant margin, yet it punches dramatically above its weight class. MiniMax M2.5 scored 80.2% on SWE-Bench Verified — matching Claude Opus 4.6 and exceeding GPT-5.2 — while costing roughly one-twentieth of the price per token. MiniMax reportedly uses M2.5 internally for 30% of all company tasks, with 80% of their newly committed code generated by the model.

The model's secret weapon is MiniMax's proprietary Forge reinforcement learning framework, which trained the model across over 200,000 real-world environments in more than 10 programming languages. This approach cultivated what MiniMax calls an "Architect Mindset" — the model proactively plans structure, features, and UI design before writing a single line of code. For organizations concerned about inference costs, M2.5 presents an extraordinarily compelling proposition: near-frontier performance at a fraction of the computational cost of its larger S-tier peers.

Qwen 3.5 — Alibaba's Global Contender (February 16, 2026)

Rounding out the S-tier is Qwen 3.5 from Alibaba's Qwen team, released on February 16, 2026 — Chinese New Year's Day — at 397 billion total parameters with only 17 billion active. The Qwen series has been on a remarkable upward trajectory, and version 3.5 solidifies Alibaba's position as one of the premier open-source model providers in the world. The model introduces Gated Delta Networks that replace standard attention in 75% of layers, achieving 19x faster decoding than its trillion-parameter predecessor Qwen3-Max while costing 60% less.

Qwen 3.5 is particularly notable for its exceptional multilingual capabilities — supporting an industry-leading 201 languages and dialects — and its native visual agent capabilities, allowing it to operate across mobile and desktop applications autonomously. Released under the permissive Apache 2.0 license with a hosted Qwen-3.5-Plus variant offering a 1-million-token context window, it gives developers maximum flexibility for both self-hosted and cloud deployments.

A-Tier: Serious Contenders with Distinctive Strengths

The A-tier is where the competition becomes truly fierce. Six models populate this tier, spanning releases from mid-2025 through early 2026: **DeepSeek R1 (671B,**

January 2025), **Mistral Large 3 (675B, December 2025**), **GPT-oss 120B (117B, August 2025**), **DeepSeek V3.2 (685B, December 2025**), **Step-3.5 Flash (196B**), and **MiMo-V2-Flash (309B)**. These models are not far behind the S-tier leaders and, in specific domains, may even surpass them.

DeepSeek's Dual Presence (January & December 2025)

DeepSeek continues to be one of the most exciting players in the open-source LLM space. DeepSeek R1, launched on January 20, 2025, was the model that truly shook the world — briefly surpassing ChatGPT as the most-downloaded app on the iOS App Store and triggering an $600 billion single-day drop in NVIDIA's market cap. At 671 billion parameters, R1 pioneered extended chain-of-thought reasoning in open-source contexts and spawned an entire generation of distilled variants that appear across C and D tiers.

DeepSeek V3.2, released on December 1, 2025, takes a different approach as a more general-purpose model with 685 billion parameters. It introduced DeepSeek Sparse Attention for efficient long-context handling, agent training across 1,800+ environments, and was the first DeepSeek model to integrate thinking directly into tool-use. The interplay between these two models gives the DeepSeek ecosystem unique versatility — developers can choose R1 for deep analytical reasoning or V3.2 for a well-rounded generalist.

Mistral Large 3 — Europe's Flagship (December 2, 2025)

Mistral Large 3 from the French AI lab Mistral AI, launched on December 2, 2025, represents Europe's strongest and most significant showing on the entire list. At 675 billion parameters with 41 billion active, it is Mistral's first MoE model since the seminal Mixtral series. Trained from scratch on 3,000 NVIDIA H200 GPUs, it demonstrates definitively that European AI research can compete at the highest international levels. Released under the Apache 2.0 license, it supports a 256K context window and has earned the second-highest ranking among open models on LMArena.

GPT-oss 120B — OpenAI Enters the Arena (August 5, 2025)

The release of GPT-oss on August 5, 2025 marked a historic moment: OpenAI's first fully open-weight language models since GPT-2. Available under the Apache 2.0 license, GPT-oss 120B uses a MoE architecture with only 5.1 billion active parameters per token, enabling it to run on a single 80GB GPU. It matches or exceeds OpenAI's own o4-mini on core reasoning benchmarks, scoring 94.2% on

MMLU and 96.6% on AIME math. Sam Altman had acknowledged months earlier that OpenAI was “on the wrong side of history” regarding open source — GPT-oss was the tangible correction.

The Flash Models — Speed Meets Quality (Early 2026)

Step-3.5 Flash (196B) and MiMo-V2-Flash (309B) round out the A-tier with an emphasis on inference speed alongside benchmark performance. The “Flash” designation signals models specifically optimized for low-latency inference — using speculative decoding, quantization-aware training, and architecture modifications that reduce time-to-first-token without sacrificing output quality. For production applications where user experience depends on response speed, these Flash models may be more practically valuable than larger, slower models ranked above them.

B-Tier: The Production Workhorses

The B-tier houses models that represent the optimal cost-performance sweet spot for real-world deployments. Six models reside here, spanning releases from March to August 2025: **Llama 4 Maverick (400B, April 2025)**, **Nemotron Ultra 253B (March 2025)**, **Qwen3–235B-A22B (April 2025)**, **Hunyuan 2.0 (406B)**, **GPT-oss 20B (August 2025)**, and **Llama 4 Scout (109B, April 2025)**.

Meta’s Llama 4 — The Ecosystem King (April 5, 2025)

Meta’s Llama 4 Maverick and Llama 4 Scout, both released on April 5, 2025, remain among the most widely recognized names in the open-source LLM ecosystem. Maverick’s B-tier placement may surprise some observers, but it reflects the increasingly competitive nature of the field rather than any fundamental weakness. What Llama 4 lacks in top-tier benchmark scores, it compensates for with an unparalleled ecosystem: thousands of fine-tuned variants, comprehensive tooling, integration with virtually every ML framework, and extensive community documentation. For many teams, these ecosystem advantages outweigh marginal performance differences.

NVIDIA’s Nemotron Ultra (March 18, 2025) & Qwen3 MoE (April 29, 2025)

NVIDIA’s Nemotron Ultra 253B, released March 18, 2025, carries strategic significance as the GPU maker’s entry into the model training space — designed to work optimally on NVIDIA’s own hardware ecosystem. Qwen3–235B-A22B, launched April 29, 2025, uses MoE with 235 billion total parameters but only 22 billion active, achieving B-tier performance while consuming inference resources comparable to a

22B dense model. Tencent's Hunyuan 2.0 at 406B and OpenAI's GPT-oss 20B (August 5, 2025) — which fits on a device with just 16GB of memory — complete the tier.

C-Tier: The Capable Middle Ground

The C-tier serves organizations with limited computational resources or specific narrow use cases. Seven models populate this tier, with some dating back to late 2024: **Llama 3.3 70B (December 2024)**, **DS-R1-Distill-Llama-70B (January 2025)**, **Qwen 2.5–72B (September 2024)**, **Gemma 3 27B (March 2025)**, **DS-R1-Distill-Qwen-32B (January 2025)**, **Command R+ (April 2024)**, and **Qwen2.5-Coder-32B (November 2024)**.

The Rise of Reasoning Distillation (January 2025)

The multiple DeepSeek R1 distilled models in this tier — all released alongside the original R1 on January 20, 2025 — illustrate one of the most consequential trends in AI: knowledge distillation from larger reasoning models into smaller, deployable ones. DS-R1-Distill-Llama-70B and DS-R1-Distill-Qwen-32B transfer chain-of-thought reasoning capabilities of the massive 671B DeepSeek R1 into models that can run on a single workstation. The distillation process trains smaller models to replicate not just final outputs but intermediate reasoning steps, transferring a surprising amount of capability even across significant parameter count reductions.

Gemma 3 27B (March 12, 2025) & Domain Specialists

Gemma 3 27B from Google, launched March 12, 2025, stands out as remarkably efficient — competing with models two to three times its size and runnable on a single consumer GPU with 24GB VRAM. Qwen2.5-Coder-32B (November 2024) is purpose-built for code generation, while Command R+ from Cohere (April 2024) optimizes for retrieval-augmented generation workflows. These specialized models often outperform higher-ranked generalists in their respective domains, illustrating the growing importance of task-specific optimization.

D-Tier: Small Models, Big Impact

The D-tier should emphatically not be dismissed. These models represent the frontier of efficient AI and are often the ones that actually get deployed at massive scale. They include: **Mistral Small 3.1 (24B, March 2025)**, **Phi-4 (14B, February 2025)**, **Llama 3.1–8B (July 2024)**, **Qwen3–30B-A3B (April 2025)**, **Gemma 3 12B (March 2025)**, **DS-R1-Distill-Qwen-14B/7B (January 2025)**, and **Phi-4-mini (3.8B, February 2025)**.

Microsoft's Phi Series (February 2025)

Microsoft's Phi-4 (14B) and Phi-4-mini (3.8B), both launched February 26, 2025, continue to redefine what small models can accomplish. Microsoft's approach centers on training with exceptionally high-quality, carefully curated data rather than simply scaling parameters. Phi-4-mini at 3.8 billion parameters is small enough to run on modern smartphones, enabling AI applications previously impossible without cloud connectivity: offline assistants, on-device code completion, and real-time translation in environments with no internet access. As mobile hardware continues to improve and quantization techniques become more sophisticated, models in this class will become the backbone of ubiquitous, always-available AI.

Qwen3–30B-A3B (April 2025) represents extreme MoE sparsity — 30 billion total parameters with only 3 billion active — making it computationally equivalent to running a 3B model while carrying the knowledge of a 30B system. The distilled DeepSeek reasoning models at 14B and 7B (January 2025) push the boundaries of compact reasoning, while Meta's venerable Llama 3.1–8B (July 2024) remains one of the most widely deployed open-source models globally thanks to its age, stability, and extensive community support.

Key Takeaways and Industry Trends

1. Chinese AI Labs Dominate the Leaderboard

The most striking observation is the overwhelming presence of Chinese AI organizations. Three of four S-tier models are Chinese. Kimi, GLM, MiniMax, Qwen, DeepSeek, Hunyuan, Step, and MiMo have collectively placed models in every single tier from S to D. This comprehensive dominance challenges the narrative that Western labs hold a decisive lead and underscores China's strategic commitment to AI supremacy — even as U.S. chip export controls attempt to limit their access to advanced hardware. GLM-5's training entirely on Huawei Ascend chips without a single NVIDIA GPU is perhaps the most geopolitically significant data point of all.

2. The February 2026 "Model Week" Was Unprecedented

The launch dates tell an extraordinary story. Between February 11 and February 16, 2026, three of the four S-tier models were released in rapid succession — GLM-5 on February 11, MiniMax M2.5 on February 12, and Qwen 3.5 on February 16 — all timed around the Chinese New Year holiday. Combined with Kimi K2.5's January 27 release, this three-week period represented the single most concentrated burst of frontier open-source AI releases in history. The strategic timing — each lab racing to

establish positioning before and during the holiday — mirrors the competitive dynamics that have come to define Chinese AI.

3. Scale Is Necessary but Not Sufficient

MiniMax M2.5's S-tier performance at 230B total (10B active) parameters, while multiple 400B+ models sit in B-tier, proves that architecture and training methodology can overcome substantial scale disadvantages. Gemma 3 27B competing with 70B models reinforces this lesson. The models achieving the best results per parameter are those combining MoE sparsity, novel attention mechanisms, and rigorously curated training data.

4. Reasoning Distillation Is Democratizing Advanced AI

The proliferation of DeepSeek R1 distilled variants at 70B, 32B, 14B, and 7B sizes — all released on the same day as R1 itself in January 2025 — highlights a powerful paradigm. A large, expensive-to-train reasoning model serves as a teacher for smaller student models, allowing organizations with modest budgets to benefit from frontier-scale research. This creates a new economic dynamic: the most valuable contribution may not be a model itself, but one that serves as an effective teacher for distillation.

5. The MoE Architecture Is Winning

The dominance of Mixture of Experts across all tiers is unmistakable. Every S-tier model uses MoE. Kimi K2.5 activates 32B of 1T, Qwen 3.5 activates 17B of 397B, MiniMax M2.5 activates 10B of 230B, and GLM-5 activates 40B of 744B. The trend extends to A, B, and D tiers. As serving costs become the primary concern for production deployments, MoE's ability to deliver knowledge capacity far exceeding active compute requirements makes it the clear architecture of the future.

6. The Proprietary-Open Gap Has Collapsed

Perhaps the most consequential takeaway: the quality gap between the best open-source and proprietary systems has effectively collapsed for most practical purposes. S-tier open models compete directly with GPT-5.2, Claude Opus 4.5/4.6, and Gemini 3 Pro on standard benchmarks. Organizations can now self-host comparable quality, maintaining full control over data, customizing through fine-tuning, and eliminating per-token API costs. This shift is particularly significant for enterprises in regulated industries where data sovereignty concerns have historically limited AI adoption.

Conclusion: The Golden Age of Open-Source AI

This tier list captures a moment of extraordinary vibrancy in the open-source AI ecosystem. From trillion-parameter behemoths to 3.8 billion parameter models compact enough to run on a smartphone, the range and quality of available options has never been greater. The concentrated burst of frontier releases in early 2026 — with four S-tier models launching within three weeks of each other — demonstrates an industry operating at a pace of innovation that would have seemed implausible just two years ago.

For practitioners choosing a model, the decision now depends less on whether a capable open-source option exists — it almost certainly does — and more on the specific constraints of their deployment. Those with substantial compute can deploy S-tier models rivaling the best proprietary offerings. The B and C-tier models provide the practical sweet spot for most production deployments. And the D-tier models are quietly revolutionizing edge AI and on-device inference, bringing intelligent computation to environments previously unreachable.

The timeline itself tells the story of acceleration. From DeepSeek R1's world-shaking debut in January 2025 to the February 2026 model blitz, the open-source community has compressed years of progress into months. Labs are learning from each other's innovations — distillation techniques, MoE architectures, sparse attention mechanisms, reinforcement learning frameworks — and the resulting competition benefits everyone. One thing is certain: the open-source LLM movement is no longer playing catch-up. It is setting the pace for the en

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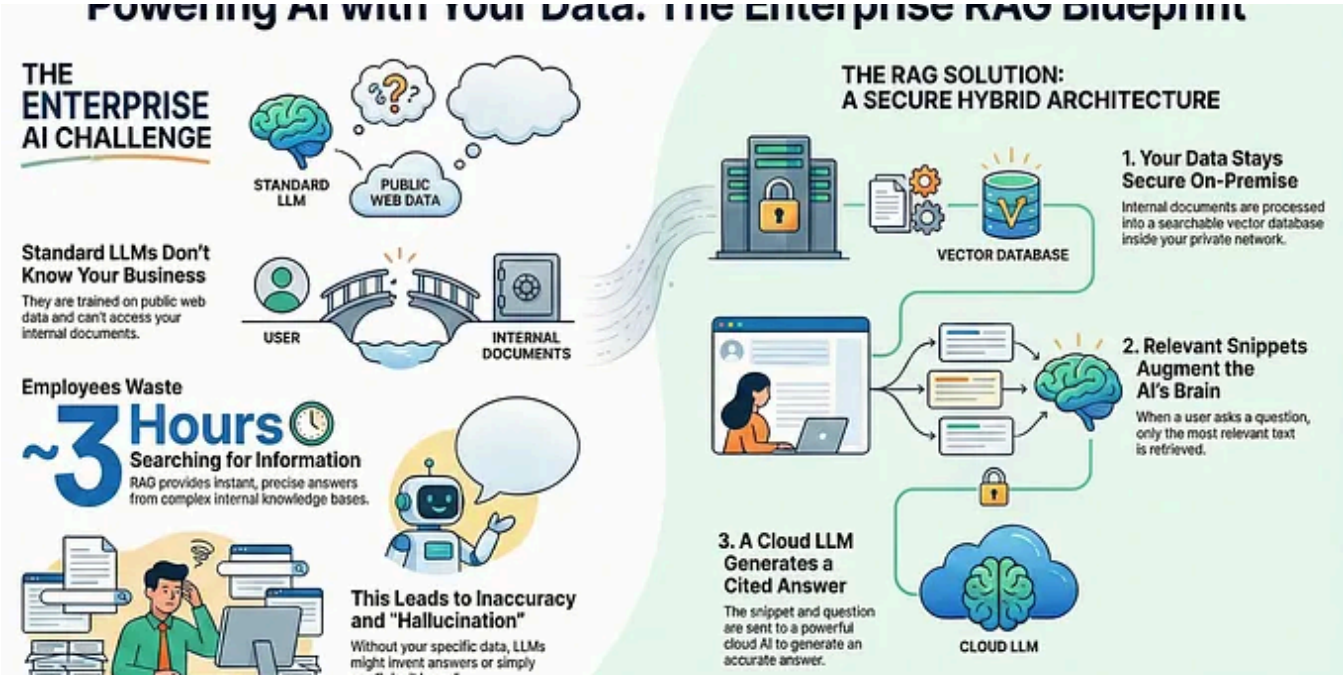

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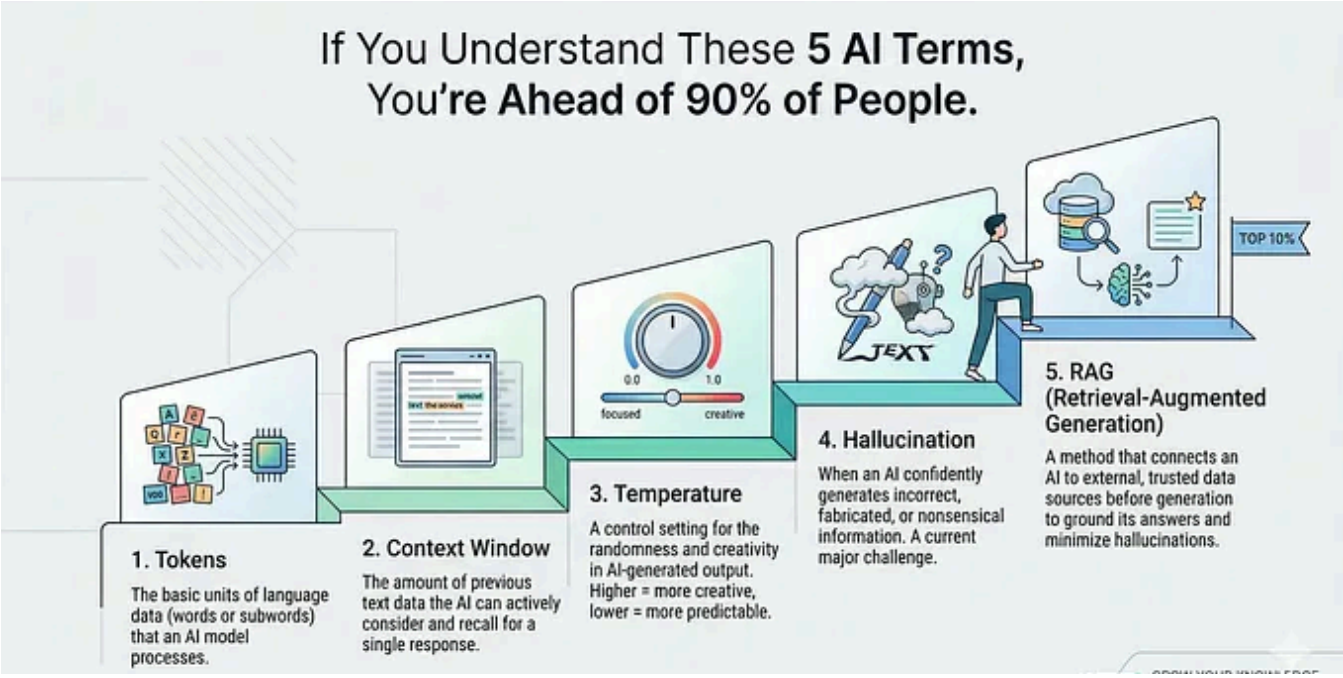

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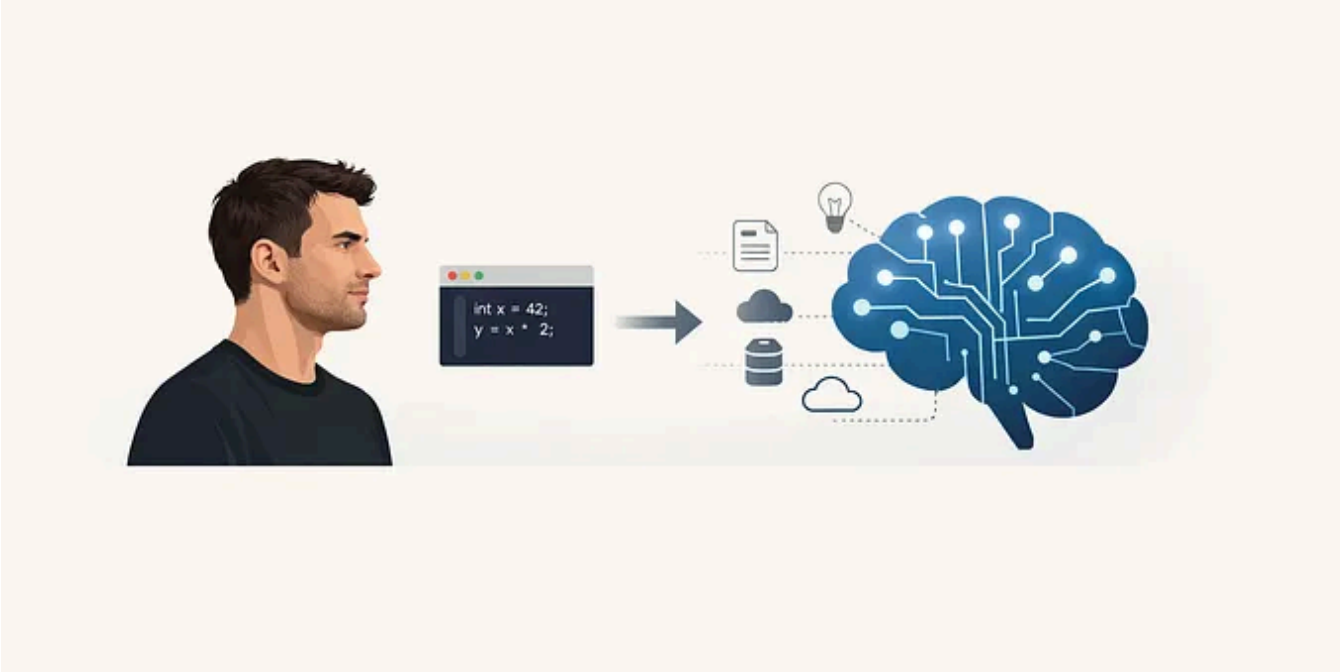

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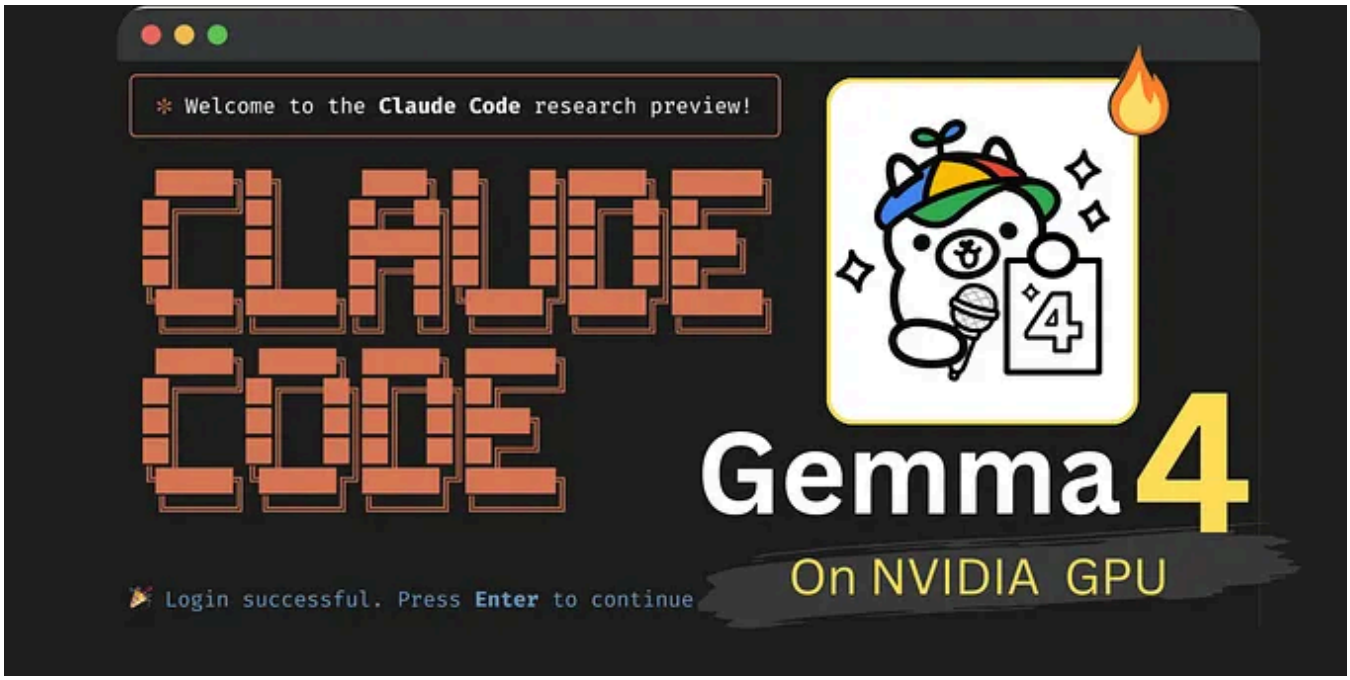

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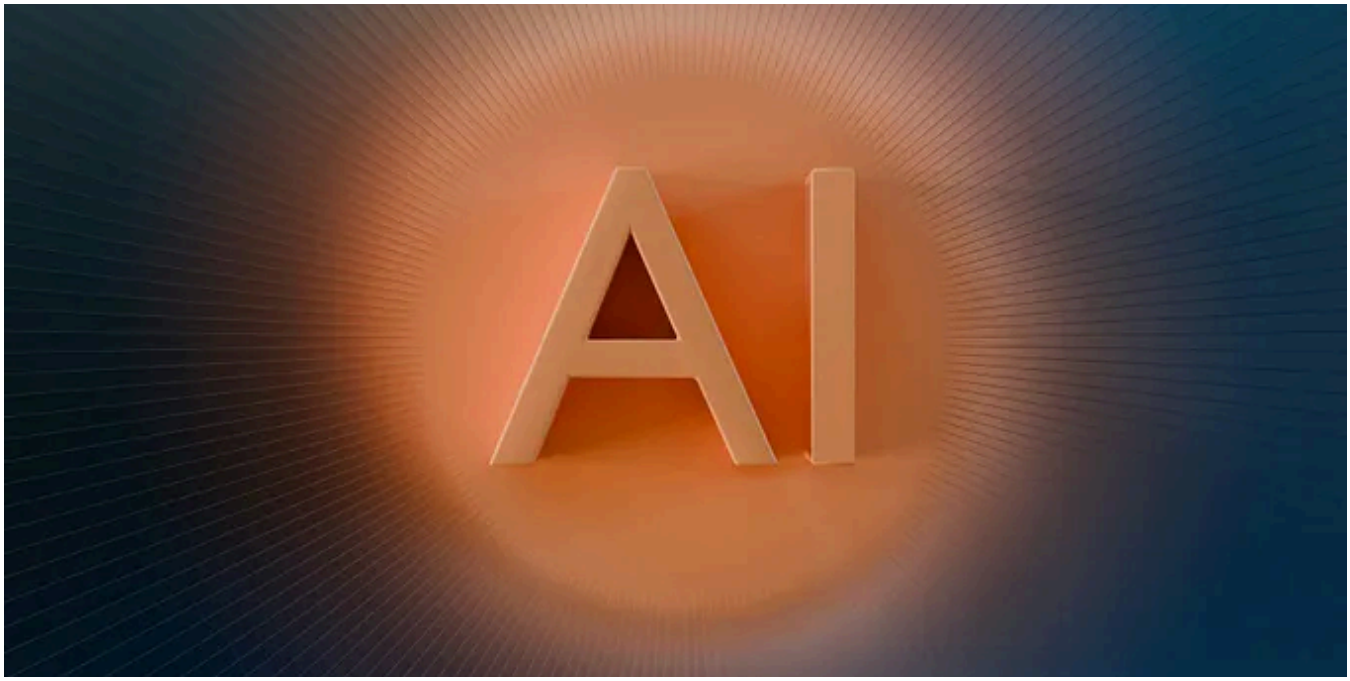

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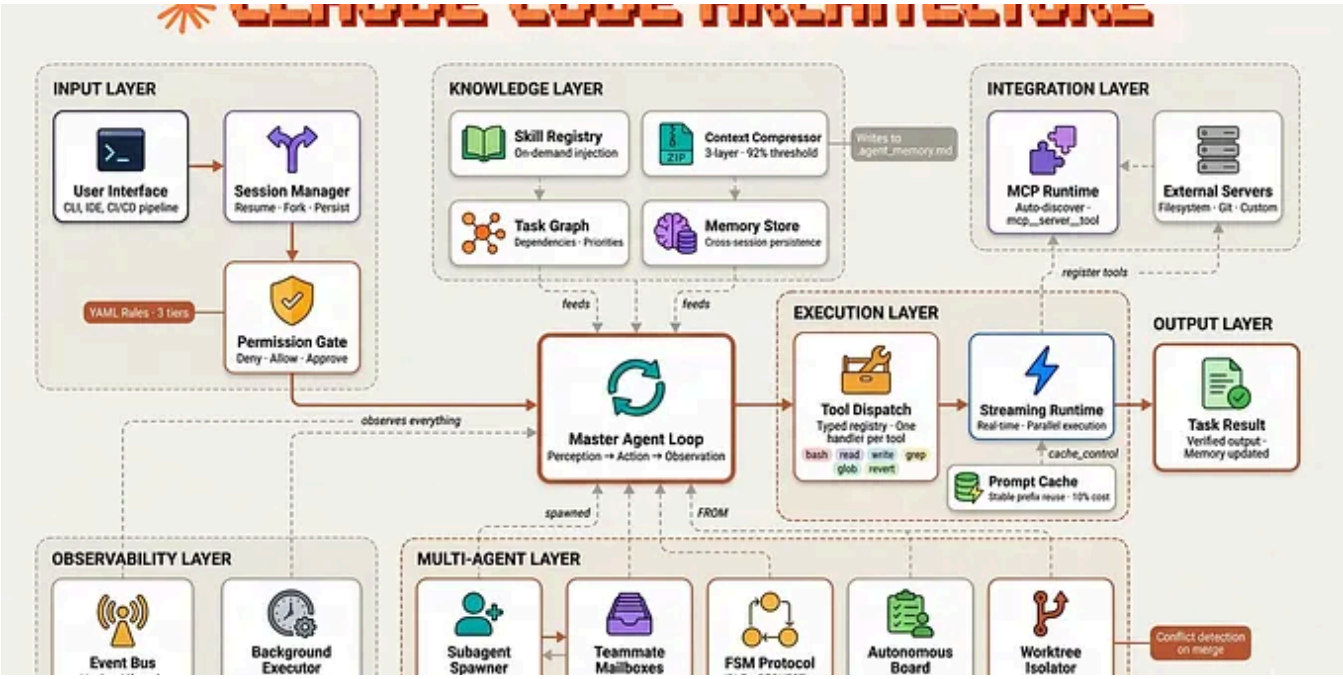

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